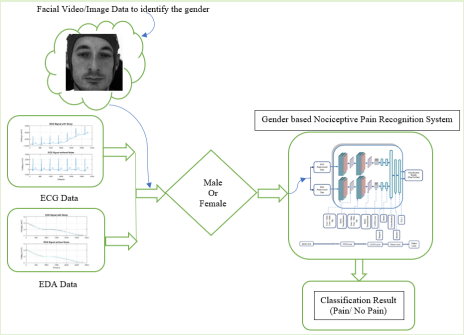
Automated Nociceptive Pain Assessment Using Physiological Signals and a Hybrid Deep Learning Network

Saranya Devi Subramaniam and Brindha Dass

***Abstract*—Postoperative pain management has become a major medical and nursing challenge. Nowadays, hospitals have taken initiatives to measure acute pain using self-report measures like the Visual Analogue Scale and Numeric Pain Intensity Scale. But these methods are inaccurate as it depends on patient's input. Therefore, there is a requirement for an objective, quantitative method to monitor pain continuously. In this work, an automated acute nociceptive pain recognition system was proposed to objectively measure nociceptive pain using physiological signals and a hybrid Deep Learning network. The hybrid deep learning network constitutes a shallow CNN network that extracts the essential information of the pain from the physiological signals and the extracted feature matrix is fed to the LSTM network for feature concatenation. This process realizes the mapping of nociceptive pain from input data to detection. This work utilizes the BioVid Heat Pain database. A Unimodal hybrid CNN_LSTM network (using ECG signals) has achieved 68.70 percent, 62.61 percent, 67.86 percent, and 75.21 percent of classification accuracy for classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4). Similarly, for classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4), the unimodal hybrid CNN_LSTM network (using EDA signals) achieved 85.65 percent, 74.47 percent, 80.80 percent, and 80.17 percent of classification accuracy. Finally, for classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4), the CNN_LSTM multimodal hybrid network (using both ECG and EDA signals) achieved 93.91 percent, 86.97 percent, 90.75 percent, and 94.12 percent of the classification accuracy, respectively.**



***Index Terms*—Biomedical signal processing, electrocardiography, electrodermal activity, deep learning networks, pain intensity classification.**

I. Introduction

PAIN is the dominant reason for any individuals to appeal for medical attention. The word pain is learned by every individual through experience. Research results of [1] show that 52.2% of patients complain about their pain as a major problem and only 34.1% complaints about other problems. Author [2] research results summarize that approx. 35% of hospitalized patients are suffering from severe acute pain and approx. 50% of patients are experiencing mild pain. In general, if the pain is not treated when necessary then the person's ordinary life is being affected straightforwardly or by implication by trading off immune, nervous and endocrine system's function.

Pain might be affected by factors like sexual orientation, physical status, conduct, emotional status, pain history, and intellectual hindrance. By considering these elements appropriate pain appraisal model must be planned [3]. An Individual's pain experience can be diagnosed using parameters such as pain cause (the reason for the pain), pain response (verbal or non-verbal communication), and pain assessment (caretaker). In most cases, pain cause and pain response can be easily identified but rarely it fails. So, the pain assessment task has to be executed properly to determine acute pain.

To manage pain at hospitals, the concept of Acute Pain Services (APS) has been emerged. Acute Pain Services (APS) is to "dedicate healthcare professional to assess, monitor, and treat pain in the Intensive Care Unit". APS is also responsible for treating side effects of patients due to pain medication daily. The investigation will be performed regularly to analyze the effectiveness of the treatment. APS is followed

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in developed countries for the past three decades with proper pain protocols and guidelines. In India, APS is still ignored and pending in day to day medical practices [4]. Data from [5] has confirmed that the approach of APS has not been evolved in India properly. Only surgeons and anesthesiologists are treating pain in the operation theatre and intensive care unit.

Measuring acute nociceptive pain helps to understand the adverse situations taking place inside the body such as tissue damage [6]. Acute pain is short-lived and difficult to measure because of its complexity and subjective nature. At present, acute pain assessment task is implemented by clinicians/caretakers by using self-report or pain scale measures. In any case, these procedures are not effective to observe pain in patients having psychological disorders and require profoundly talented spectators to quantify pain. Utilizing these procedures, it is additionally hard to pick the pain-relieving drug doses to the patients in the emergency unit. Despite its influence, acute pain is often underestimated and untreated [7]. Therefore, a valid and reliable automated computer-based system is required to detect and monitor patient pain.

According to the theories, it is claimed that pain is related to autonomic dysfunction [8]. The Autonomic Nervous System (ANS) helps to feel the severity of pain. Its action is unconsciously controlled by human brainstem. It has two divisions; they are the Parasympathetic Nervous System (PNS) and Sympathetic Nervous System (SNS). The PNS is activated during rest and the SNS is activated during stress/pain. At present, clinicians are using self-report measures such as the Numeric Rating Scale (NRS), Visual Analogue Scale (VAS), etc. to determine pain. But these measures are not useful as they are subjective and not stable [3]. The ANS activates the SNS and causes prominent changes in the physiological signals such as Electrocardiography (ECG), Heart Rate Variability (HRV), Electrodermal Activity (EDA), Heart Rate (HR), etc., [9] which can directly be used to determine the pain levels.

Thus, to achieve this goal many researchers have recently focused their research on physiological signals (ECG, Electromyogram (EMG), Photoplethysmography (PPG), and EDA to determine and monitor acute pain levels [10].

A. Related Works and Motivation

Extensive research has been performed in the area of the acute adult pain recognition system. Below are the few articles which support the proposed work. The author Cho D. *et al.*, [11] had developed a physiological signal based stress level measuring system. In this work, a virtual reality device is used to generate five different levels of stress (baseline, mild stress, moderate stress, severe stress, and recovery). Physiological signals such as PPG, peak characteristics of EDA, and changes in Skin Temperature (SKT) are used to quantitatively determine the stress level. The features from all the above-mentioned signals are extracted and found statistically significant parameters using an ANOVA test. Finally, these features are given as input to the Extreme Learning Machine (ELM) classifier to classify the five different stress levels. It has achieved 95% classification accuracy.

Daniel Lopez-Martinez and Rosalind Picard [12] have presented a Long Short-Term Memory Neural Network (LSTM-NN) algorithm to deconvolve the EDA signal into tonic and phasic components and then extracting the features from it to feed the data to LSTM recurrent neural network regressor which estimates the different levels of pain. The author has also utilized Heart Rate Variability to perform pain estimation task, but not outperformed the EDA results. Yaqi chu *et al.* [6] used BVP, ECG, and SCL to determine four levels of pain. The author has developed a hybrid genetic algorithm to detect different levels of pain for distinct schemes which include single-signal, multi-subject, multi-day, between-subject, and multi-signal.

In recent years, deep learning approaches [13]–[19] are widely used in many applications related to physiological signals. In the process of affective modeling, difficult phases are feature extraction and feature selection. This problem can be resolved by implying Deep Learning (DL) approaches. The following are the research articles that explain the use of DL in the recognition of different pain levels. L. Santamaria-Granados *et al.* [20] have developed an emotion detection model using Deep Convolutional Neural Network (DCNN). This model was efficient to determine the classification of arousal and valence states using ECG and EDA signals. Patrick Thaim *et al.* [21] has utilized EDA, ECG, and EMG signals and designed the CNN network to classify pain Vs no pain condition with the classification accuracy of 84.57%. The author has proved that the EDA is providing promising results to classify Pain Vs No pain better than other physiological signals. From all the above studies, it is evident that the physiological signals ECG and EDA can be used for pain recognition studies. These articles prove that Deep Learning algorithms (DL) can outperform the standard algorithms which utilize handcrafted features.

B. Objective and Key Contribution

The main objective of this article is to propose a shallow DL algorithm to determine acute nociceptive pain. The key contributions of this manuscript are:

- An inspection of the use of physiological signals such as EDA and ECG signals for determining nociceptive pain.
- An exploration of preprocessing steps to remove unwanted noises from ECG and EDA signals.
- An investigation of the effectiveness of the use of a shallow hybrid CNN-LSTM network for the determination of nociceptive pain using physiological signals such as ECG and EDA.
- This study also explores the gender influence in the determination of acute nociceptive pain.

Evidence from several cohort studies indicated that the ECG and EDA signals are the best sources of the presence of acute pain in adults. Therefore, these signals are utilized to determine the nociceptive pain using the shallow hybrid CNN-LSTM network. The performance of this network is mainly based on the selection of the hyperparameters. Discrete methods are being tried out to fix the suitable values for the parameters. Fig. 1 illustrates the overview of the proposed work. Here, the proposed network has five layers they are

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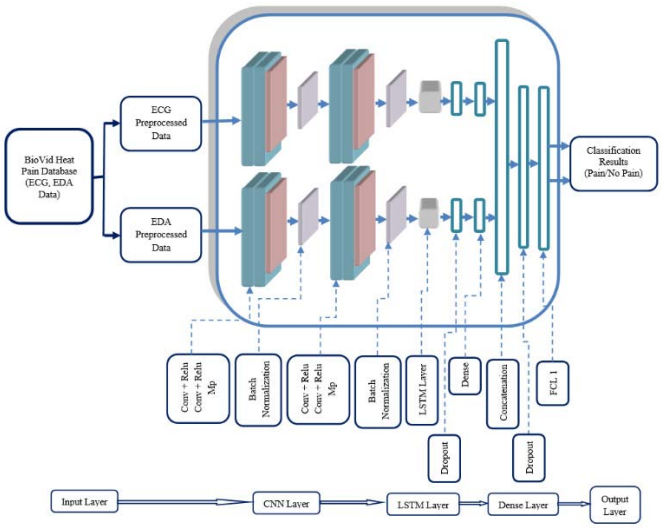


Fig. 1. Overview of the proposed system. ∗ Conv – Convolution layer; ReLU – Rectified Linear Unit; Mp – Maxpooling layer; LSTM – Long Short-Term Memory Network; FCL – Fully Connected Layer.

input layer, the CNN layer, the LSTM layer, the dense layer, and an output layer.

II. Approach and Methodology

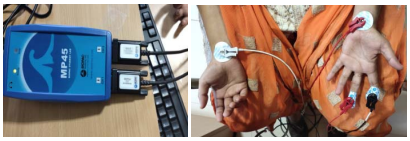
Sections such as the dataset used for the analysis, pre-processing steps involved, and finally, the deep learning algorithm built for measuring pain are discussed in detail in the following subsections. When compared to the conventional machine learning algorithms, deep learning algorithms are chosen for the study, as they provide good classification results without the use of man-made features.

A. BioVid Heat Pain Database

The BioVid Heat Pain database [22] is a publicly available integrated dataset with bio-potential signals (ECG, GSR, and EMG at Trapezius muscle) and facial action video signals. This dataset is collected in collaboration with the Neuro-Information Technology group of the University of Magdeburg and the Medical Psychology group of the University of Ulm. The data are collected by subjecting 4 levels of heat stimulus (BL1 (Baseline Level) – No pain, PA1 (Pain Level 1) – Mild Pain, PA2 (Pain Level 2) – Moderate pain, PA3 (Pain Level 3) – Severe pain, PA4 (Pain Level 4) – Very severe pain) to 87 healthy participants under a controlled environment. Each stimulus is subjected to 4 seconds to participants in an arbitrary manner for 20 times and provided a resting time of 8 – 12 seconds between each pain stimulus. In this study, 67 subjects of the Part-A BioVid Heat Pain Database are used. Each subject has 100 recordings of both video and bio-signals with proper VAS labels.

B. Experimental Setup and Real-Time Data Collection

The real-time data collection was carried out in this work using the Biopac setup. Biopac Student Lab 4.0 software is used for data analysis, data visualization, and interpretation.



(a) (b)

Fig. 2. (a) Biopac MP45 data acquisition unit (b) ECG and EDA Electrode placement.

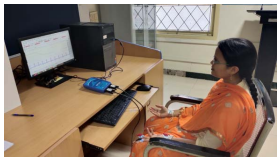


Fig. 3. Experimental setup for acquiring EDA and ECG signals in real-time.

Where the Biopac MP45 hardware system is used for data acquisition purposes. Biopac MP45 is a two-channel data acquisition unit where channel I is connected to the SS57LA electrode lead to collect the EDA signal in real-time, and channel II is connected to the SS2LB Electrode lead to collect the ECG signals in real-time from normal subjects (data collected only in the No pain condition).

In this study, 10 normal physiological data of the subject are collected to test the performance of the nociceptive pain recognition system. Figure Fig. 2(a) presents the MP45 set-up for the acquisition of real-time data and figure Fig. 2(b) shows the positioning of the electrodes for the acquisition of bio-signals. Normal surface electrodes are used for the collection of ECG data and EDA isotonic gel snap electrodes are used for the collection of EDA data. Figure Fig. 3 represents the complete setup (Biopac student lab 4.0 and Biopac MP45 hardware) for the acquisition of real-time data.

C. Physiological Data Acquisition

To obtain an ECG and EDA signal from Biopac MP45, the sampling rate is fixed at 2000 samples/sec and the gain is fixed at $\times 1000$. The recorded ECG and EDA signals may be affected by artifacts and noise. Appropriate filtering methods are therefore applied to each modality. ECG signals may be affected by baseline low-frequency noise (less than 0.5 Hz) generated by patients breathing and may also be affected by high-frequency noise (greater than 40 Hz). The third-order Bandpass Butterworth filter with a frequency length of [0.1, 40] Hz is therefore used to remove unwanted noise from ECG signals. Figure Fig. 4(a) shows the ECG signal with and without noise. EDA signals may be affected by motion artifacts and other noise components, therefore a third-order low-pass filter with a cutoff frequency of 0.2Hz was used to filter noise without affecting the slowly varying EDA signal data. Figure Fig. 4(b) shows the EDA signal with and without noise. The classification accuracy of the proposed method is highly dependent on the noise removal process of the physiological signals.

Visualizing figure Fig. 4, it is prompt that the unwanted noise of the physiological signal has been successfully

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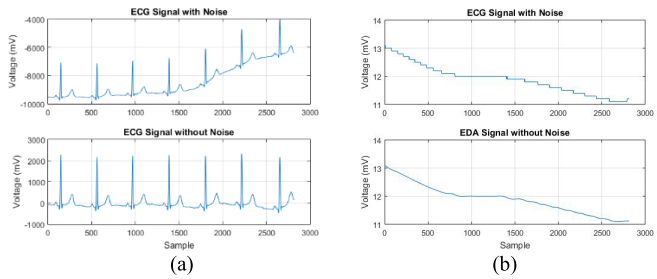

Fig. 4. (a) Real-time ECG Signal with and without noise (b) Real-time EDA signal with and without noise.

removed. Without the noise removal process, the classification accuracy of the proposed system is affected by over-fitting.

D. Data Analysis

The present work has considered 67 subject's physiological data (EDA and ECG) from Part-A filtered BioVid Heat Pain Database. For 27 subjects, 14 files are chosen out of 20 records of physiological data for processing, and for the other 40 subjects whole of 20 records of bio-signals data are chosen for the study. Therefore, in total ECG data is represented as (40 subjects X 20 records X 5 different pain levels) + (27 subjects X 14 records X 5 different pain levels) = 5890 sequences of time series data. This method of data collection is used for the study to just reduce the computation complexity. In the same way of representation, EDA data also have 5890 sequences of time series data. All the data is labeled with corresponding pain levels using VAS.

E. A Deep Learning Model Based on a Hybrid Network: CNN_LSTM

The methods used in this work are described in detail in this section. CNN is one of the most popularly used Multilayer Perceptron Networks, designed to effectively identify various patterns, including transformations, variations, and geometric distortions.

In this work, CNN takes its input as one-dimensional sequential time series data. The proposed hybrid CNN_LSTM architecture consists of an Input Layer (IL), Convolutional Layer (CL), Pooling Layer (PL), Batch Normalization layer (BNL), LSTM Layer, and a Fully Connected Layer (FCL). In the convolutional layer, the input and the filter are convolved to form feature maps. All the units of the feature maps share the same weights. This layer is impacted by the kernel dimension. The non-linear activation function used in this work is Rectified Linear Units (ReLUs). It initiates better propagation and acceleration of the training convergence using stochastic gradient descent. The pooling layer reduces the dimensionality by half without losing the depth information. It favors the most significant spatial features and feeds the data to the LSTM layer.

As discussed previously, CNN layers are excellent for extracting the spatial features of physiological signals, and the successor LSTM layer is responsible for capturing the temporal dynamics of the feature maps. The LSTM layer fuses similar data and can filter out the unwanted information extracted by the CNN algorithm. This process helps to store the important features in the memory unit for a long time. The output of this layer is fed to the Fully Connected Layer (FCL). This layer contains the Sigmoid function which contributes to calculating the probability distribution over each class. Thus, FCL will estimate the class which will be the final result of the CNN.

Overfitting of the CNN_LSTM hybrid model may occur during the training phase. By introducing the concept of a drop-out regularization method in the network, it prevents neurons from being extremely well adapted to training data [23]. As explained in figure Fig. 1. CNN is used to perform Feature Extraction (FE) and the LSTM is used to fuse and store the important features to the output stage. In the FE stage as in the equation (1), the output $z_i^{(l)}$ of l layer consists of m_1^l feature maps, $b_i^{(l)}$ bias matrix and $k_{i,j}^{(l)}$ filters.

$$Z_i^l = tanh\left(b_i^l + pooling_{max}\left(\sum_{j=1}^{m_1^{l-i}} k_{i,j}^l * z_j^{l-1}\right)\right) \tag{1}$$

The working of the LSTM layers is mathematically represented as follows:

$$fs_t = \sigma\left(\left(w_{hfs}.h_{t-1}\right) + \left(w_{gfs}.g_t\right) + b_{fs}\right) \tag{2}$$
$$in_t = \sigma\left(\left(w_{hin}.h_{t-1}\right) + \left(w_{gin}.g_t\right) + b_{in}\right) \tag{3}$$
$$q_t = \sigma\left(\left(w_{hq}.h_{t-1}\right) + \left(w_{gq}.g_t\right) + b_q\right) \tag{4}$$
$$\widehat{cl}_t = tanh\left(\left(w_{hcl}.h_{t-1}\right) + \left(w_{gcl}.g_t\right) + b_{cl}\right) \tag{5}$$
$$cl_t = cl_{t-1} * fs_t + in_t * \widehat{cl}_t \tag{6}$$
$$h_t = q_t * tanh\left(cl_t\right) \tag{7}$$

Here, g_t is the input to the LSTM network cl_t in equation (7) refers to long-term memory, $\sigma(*)$ refers to Sigmoid function, if the output of this function is 1 then the data is fully remembered and if the output is 0 then the data is completely forgotten. To perform the above process, h_{t-1}, g_t are monitored. The forget gate fs_t in the equation (2) decide about residing the past information in the long-term memory cl_{t-1}. The input gate in_t in the equation (3) decide on how much input information can be added to the cl_t. The output gate q_t in the equation (4) decides on how much information should be available as output. w_{gfs}, w_{gin}, w_{gq} are the weights between the hidden layer and the forget gate, input gate, and the output gate respectively. The "$*$" operation corresponds to matrix multiplication and "." corresponds to multiplication between elements. The output of the LSTM network includes (cl_t) present unit state and (h_t) present output state. (h_t) in the equation (7) are the overall output of the LSTM network, which will be the input of the sigmoid function. The result of this function will be fed as input to the FCL layer of CNN to determine final classification results. The CNN_LSTM network is a hybrid deep learning network and hence it is trained with Backpropagation Through Time concept.

III. Results and Discussions

Formerly, the required pre-processing steps have to be implemented in the classification process. Here, the sampling

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rate is fixed at 512Hz and, as previously discussed in the Real-Time Data Acquisition Module, third-order Butterworth filters are used to remove unwanted noise from ECG and EDA signals [21]. Even after the filtering process, the ECG signal could be affected by the drift of the baseline, thus subtracting the signal with the fifth-order polynomial least-squares fit to detach the signal. The final data is prepared as it is proposed in [21], each signal extracted within the window of 5.5s to compose a 1D array of data of size $5.5 \times 512 = 2816$. Therefore, each physiological modality has $(27\times14 \times 2816\times1) + (60\times20 \times 2816\times1)$ data to fed to the classifier.

Women are more sensitive to pain than men, according to researchers. Their emotions directly affect the perception of pain more than men do. Women may also experience adverse effects, such as respiratory depression, when they have the same analgesia drug dosage as males. Thus, the treatment of men and women's pain is not the same and a detailed study is required to manage the pain in both. Therefore, a gender-based pain recognition system is also developed in this study. A hybrid deep learning network based on CNN_LSTM has been developed to determine the onset of pain using the female BVDB, male BVDB and for the entire BVDB dataset.

A. Hyper-Parameter Selection

Appropriate hyper-parameters have to be selected to develop the CNN_LSTM hybrid model. All the networks developed for the study use the Adaptive Moment Estimation (Adam) optimization algorithm, with the learning rate fixed to 10^{-5}. The input layer (training data) has ECG data of size (2816×1) and EDA data of size (2816×1). The next step is to develop the first hidden layer in which there are two convolutional layers (Conv1D), where both the Conv1D has 256 kernels with the size of 5 to convolve with the input data and the border_mode is set to 'same'. The next layer will be the max-pooling layer with the pooling window of size 2 and the padding is set to 'same'. The final layer of the first hidden layer is the batch normalization layer.

In the same way, the next hidden layer is developed with the two Convolutional layers, a max-pooling layer, and a batch normalization layer. A Preceding layer will be the LSTM layer with the hidden layers set to 200. Finally, the Fully Connected Layers FCL1 and an output layer are used to adjust the feature matrix to get the respective classification results. The classification output is obtained by using the 'sigmoid' function, which provides the result either as no pain or pain. Here, all the convolutional layers are utilizing 'ReLU' as activation function, ReLU(x) = max (0, x). To reduce the overfitting problems, the dropout method is applied after the LSTM layer. Here, the dropout parameter is fixed to 0.25. The performance of the shallow hybrid CNN-LSTM network is mainly based on the selection of the hyperparameters.

B. The Proposed Hybrid CNN-LSTM Architecture

In this section, the proposed hybrid CNN_LSTM architecture is explained in detail for classifying physiological signals as either no-pain or pain. The block diagram of the proposed system is displayed in figure Fig. 1. It consists of a data acquisition unit, a pre-processing unit followed by a module of the deep learning algorithm which determines the result of the classification.

TABLE I
GENDER-BASED HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING ECG SIGNALS

ECG Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing Accuracy Male DB	**95.79**	**68.91**	**76.89**	**75.63**	**72.69**
Testing Accuracy Female DB	**70.59**	**79.83**	**81.93**	**78.15**	**83.19**
Testing Samples	119	119	119	119	119

Here, physiological signals (EDA and ECG) from BioVid Heat Pain Database (BVDB) are used to test the performance of the algorithm. These signals are pre-processed in such a way that the data is without unwanted artifacts or noises. As discussed earlier, to get good classification results, this task must be implemented. The hybrid CNN-LSTM network is used for learning the pre-processed data and for measuring the pain onset. The performance of the proposed algorithm is tested on Part–A BioVid Heat Pain Database using the following metrics, classification accuracy (1) as in the equation (8), mean squared error - MSE (2) as in the equation (9), and Confusion Matrix (3). The Confusion Matrix provides details about the number of correct and incorrect predictions by the classifier.

$$classification_accuracy = \left(correct_predictions / total_predictions\right) * 100 \tag{8}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n} |x_i - x|^2 \tag{9}$$

The following paragraphs detail the different experiments executed, and the performance of the algorithm in determining the onset of pain. In this work, nociceptive pain determination based on gender is implemented and the results are tabulated in Table I, Table III, and Table V, below.

The ECG data were divided into training data (Female & Male BVDB - 1059 samples) and testing data (Female & Male BVDB - 119 samples) for the gender-based nociceptive pain determination. Similarly, The EDA data was also divided into training data (Female & Male BVDB - 1055 samples) and testing data (Female & Male BVDB - 123 samples) for the gender-based nociceptive pain determination. To analyze the entire physiological BVDB data for pain recognition, the ECG data were divided into training data (Entire BVDB – 2118) and testing data (Entire BVDB – 238). And the EDA data were divided into training data (Entire BVDB – 2119) and testing data (Entire BVDB – 237).

The tables Table I, Table II, Table III, Table IV, Table V, and Table VI show the results of the classification

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TABLE II
HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING ECG SIGNALS – ENTIRE BVDB

ECG Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing accuracy	**76.68**	**68.7**	**62.61**	**67.86**	**75.21**
MSE	0.2521	0.3109	0.3739	0.3235	0.2563
Testing Samples	238	238	238	238	238

TABLE III
GENDER-BASED HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING EDA SIGNALS

EDA Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing Accuracy Female DB	**94.31**	**77.24**	**78.86**	**54.1**	**83.33**
Testing Accuracy Male DB	**89.91**	**84.65**	**67.98**	**57.46**	**78.51**
Testing Samples	123	123	123	123	123

TABLE IV
HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING EDA SIGNALS – ENTIRE BVDB

EDA Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing Accuracy	**92.86**	**85.65**	**74.47**	**80.8**	**80.17**
MSE	0.0756	0.265	0.2531	0.194	0.1983
Testing Samples	237	237	237	237	237

of the nociceptive pain recognition system using a hybrid CNN_LSTM network and the physiological signals (ECG & EDA). The second column (BL1 (No Pain) Vs PA1+PA2+PA3+PA4 (Pain)) in all these tables represents the classification result of no pain versus pain condition. Here, all the pain level data are equally considered. The third column shows the classification result of no pain versus mild pain condition (BL1Vs PA1), the fourth column illustrates the no pain versus moderate pain condition (BL1 Vs PA2) classification results, the fifth column demonstrates the classification result of no pain versus severe pain condition (BL1 Vs PA3) and finally, the last column picturizes the classification results of no pain versus intolerable pain condition (BL1 Vs PA4).

For the classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4), Female BVDB based Unimodal hybrid CNN_LSTM network (using ECG signals) has achieved 79.83%, 81.93%, 78.15 and 83.19% of classification accuracy respectively. In the same way, Male BVDB based Unimodal hybrid CNN_LSTM network (using ECG signals) has achieved 68.91%, 76.89%, 75.63%, and 72.69% of classification accuracy respectively. Finally, for the Entire BVDB based Unimodal hybrid CNN_LSTM network (using ECG signals) has achieved 68.70%, 62.61%, 67.86%, and 75.21% of classification accuracy respectively.

For the classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4), Female BVDB based Unimodal hybrid CNN_LSTM network (using EDA signals) has achieved 77.24%, 78.86%, 54.10%, and 83.33% of classification accuracy respectively. Similarly, for the Male BVDB based Unimodal hybrid CNN_LSTM network (using EDA signals) has achieved 84.65%, 67.98%, 57.46%, and 78.51% of classification accuracy respectively. Finally, for the Entire BVDB based Unimodal hybrid CNN_LSTM network (using EDA signals) has achieved 85.65%, 74.47%, 80.80%, and 80.17% of classification accuracy respectively.

TABLE V
GENDER-BASED HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING ECG & EDA SIGNALS

ECG & EDA Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing Accuracy Female DB	**96.34**	**84.55**	**61.38**	**77.46**	**81.71**
Testing Accuracy Male DB	**95.8**	**93.09**	**86.13**	**78.15**	**76.79**
Testing Samples	123	123	123	123	123

TABLE VI
HYBRID CNN – LSTM CLASSIFICATION PERFORMANCE ON THE BVDB DATA USING ECG & EDA SIGNALS – ENTIRE BVDB

ECG & EDA Signals	Pain Vs No Pain	BL1 Vs PA1	BL1 Vs PA2	BL1 Vs PA3	BL1 Vs PA4
Testing Accuracy	**97.27**	**93.91**	**86.97**	**90.75**	**94.12**
MSE	0.0294	0.0677	0.1302	0.0924	0.0588
Testing Samples	238	238	238	238	238

The tables Table V and Table VI show the results of the Multi-modal nociceptive pain classification system using the hybrid CNN_LSTM deep learning network and the physiological signals (ECG & EDA). The ECG and EDA data were divided into training data (Female & Male BVDB - 1055 samples, Entire BVDB – 2119 samples) and testing data (Female BVDB - 123 samples, Male BVDB – 119 samples, Entire BVDB - 237) to develop the nociceptive pain recognition system.

For the classification events (BL1 Vs PA1, BL1 Vs PA2, BL1 Vs PA3, and BL1 Vs PA4), Female BVDB based Multimodal hybrid CNN_LSTM network (using ECG & EDA signals) has achieved 84.55%, 61.38%, 77.46%, and 81.71% of classification accuracy respectively. Similarly, Male BVDB based Multimodal hybrid CNN_LSTM network (using

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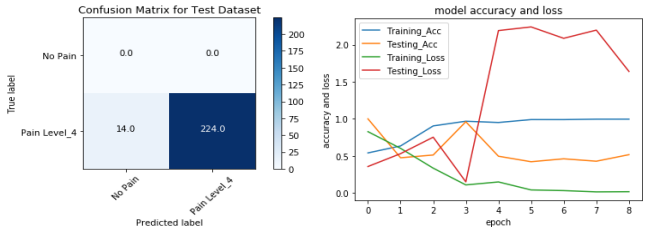

(a) (b)

Fig. 5. Multimodal Nociceptive pain recognition system results that differentiate BL1 and PA4 using Entire BVDB. (a) Confusion matrix and (b) model accuracy & loss.

TABLE VII
COMPUTATION TIME TAKEN BY THE HYBRID CNN_LSTM ALGORITHM TO DETERMINE NOCICEPTIVE PAIN

Sl. No.	Physiological Signals Entire - BVDB	Method	Computation Time
1.	ECG: **BL1 Vs PA1**	Unimodal	59.36 s
2.	ECG: **BL1 Vs PA4**	Unimodal	1 min 8.40s
3.	ECG: **BL1 Vs (PA1+PA2+PA3+PA4)**	Unimodal	2 min 21.35s
4.	EDA: **BL1 Vs PA1**	Unimodal	1 min 39.57s
5.	EDA: **BL1 Vs PA4**	Unimodal	52.44s
6.	EDA: **BL1 Vs (PA1+PA2+PA3+PA4)**	Unimodal	1 min 53.45s
7.	ECG & EDA: **BL1 Vs PA1**	Multimodal	3 min 59.13s
8.	ECG & EDA: **BL1 Vs PA4**	Multimodal	2 min 58.26s
9.	ECG & EDA: **BL1 Vs (PA1+PA2+PA3+PA4)**	Multimodal	6 min 39.29s

ECG & EDA signals) has achieved 93.09%, 86.13%, 78.15%, and 76.79% of classification accuracy respectively.

Finally, for the Entire BVDB based Multimodal hybrid CNN_LSTM network (using ECG & EDA signals) has achieved 93.91%, 86.97%, 90.75%, and 94.12% of classification accuracy respectively. The hybrid CNN_LSTM network which utilizes synchronized ECG & EDA signal is capable of determining mild pain and intolerable pain effectively when compared with the state of the art. Even the classification of no pain (BL1) and pain (PA1+PA2+PA3+PA4) is also being classified efficiently by using the same network.

Fig. 5 (a) represents the confusion matrix obtained by the Multimodal Nociceptive pain recognition system and Fig. 5 (b) displays the graphical representation of the proposed model accuracy and model loss obtained by the Multimodal Nociceptive pain recognition system, which discriminates the no pain and intolerable pain level (BL1 Vs PA4) condition using Entire BVDB data.

The computational complexity of the algorithm was documented in Table VII for both the unimodal and multimodal systems. Results reveal that our proposed network achieves significantly lower computational complexity during testing.

The performance evaluation metrics of the CNN_LSTM hybrid algorithm based on Female DB (ECG & EDA data), Male DB (ECG & EDA data), and Entire DB (ECG & EDA data) were presented in the Tables Table VIII to Table X, and their classification accuracy outperformed the other work.

TABLE VIII
RESULTS OF HYBRID CNN_LSTM ALGORITHM FOR THE CLASSIFICATION OF ENTIRE BVDB PHYSIOLOGICAL DATA INTO NO PAIN AND PAIN

Sl. No	Physio logical Signals	Methods	Classification Accuracy BL1 Vs (PA1+PA2+PA3 +PA4)	Testing Samples
1.	ECG	CNN + LSTM (without dropout regularization)	75.63 %	238
2.	ECG	CNN + dropout regularization (without LSTM layer)	74.79 %	238
3.	**ECG**	**CNN + LSTM Layer + dropout regularization**	**76.68 %**	**238**
4.	EDA	CNN + LSTM (without dropout regularization)	96.01 %	237
5.	EDA	CNN + dropout regularization (without LSTM layer)	76.26 %	237
6.	**EDA**	**CNN + LSTM Layer + dropout regularization**	**92.86 %**	**237**
7.	ECG & EDA	CNN + LSTM (without dropout regularization)	93.70 %	238
8.	ECG & EDA	CNN + dropout regularization (without LSTM layer)	80.67 %	238
9.	**ECG & EDA**	**CNN + LSTM Layer + dropout regularization**	**97.27 %**	**238**

TABLE IX
PERFORMANCE COMPARISON WITH BASELINE ALGORITHMS FOR THE CLASSIFICATION TASK BL1 VS PA4 – BVDB DATA

Method	ECG	EDA
Werner et al. [5]	62.00	73.80
M. Kächele et al. [15] [16]	53.90	81.10
Patrick Thiam et al. [28]	57.04±11.58	84.57±14.03
Our Approach	**Female DB- 83.19** **Male DB- 72.69** **Entire DB- 75.21**	**Female DB- 83.33** **Male DB- 78.51** **Entire DB- 80.17**

TABLE X
PERFORMANCE OF THE PROPOSED ALGORITHM FOR THE REAL-TIME DATA

Method	Accuracy	MSE	Test Samples
Unimodal - ECG	50	0.5	10
Unimodal - EDA	70	0.3	10
Multi-modal – ECG & EDA	40	0.6	10

Hence, it is proved that both ECG and EDA data have significant information on acute pain. Compared with the state of the art, the proposed multi-modal network using synchronized

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ECG & EDA signal is capable of effectively determining mild pain and intolerable pain. Besides, the Unimodal hybrid CNN_LSTM network has achieved good classification accuracy for both ECG & EDA signals using Female BVDB data compared to the Male BVDB and Entire BVDB. Gender-based pain recognition is always difficult, especially when determining pain in females as they are emotionally dependent. Therefore, the final results of the classification should be compared with the individuals' emotions in deciding on the patients' analgesic dosage.

Based on the data fed to the classifier, classification accuracy changes, therefore there is a need to provide a good amount of data to the classifier. The developed unimodal and multimodal nociceptive pain recognition system classifies the data either as pain or as no pain using ECG and EDA signals provide significant classification results when compared to the baseline systems.

The real-time data as tabulated in the Table. X is collected using the Biopac system and has been tested for the performance of the hybrid CNN_LSTM unimodal classification based on ECG and EDA data. Here, normal subjects between the ages of 20-30 have been chosen for the study. Only no pain data are collected from the subjects and tested the performance of the proposed algorithm.

This study contributes three modules, the first module is to develop a unimodal pain recognition system, the second module is to develop a multimodal pain recognition system and the final module is to develop a gender-based multimodal pain recognition system. All three modules have been developed successfully with acceptable classification accuracy. As mentioned previously, a gender-based multimodal pain recognition system is required to help the APS and to start the APS in India.

IV. Conclusion and Future Work

In this study, Keras and TensorFlow are used as backend with Windows Server 2012 R2 Standard enabled Deep Learning framework in Windows 2012 Operating System.

In summary, this work dealt with the use of a hybrid CNN_LSTM network for the classification of nociceptive pain based on Unimodal classification using ECG and EDA data separately and a Multimodal classification using ECG and EDA data. Even though there are few works based on a deep learning algorithm to classify pain on BVDB, the proposed work based on the unimodal classification and multimodal classification has provided significant results when compared to the state-of-the-art.

The multi-class classification has to be implemented. Considering only physiological signals to measure nociceptive pain will not be the best option as it is impacted by many parameters such as medication to the patient, fear, surgery, treatments, stress of the patients. Therefore, multimodal methods have to be implemented to quantitatively measure nociceptive pain. Even though the deep learning algorithms provide the best results, the understanding of the deep networks is complex and insufficient. Hence, the non-linear complex networks should be remodeled to linear form networks by computing eigenvalues and eigenvectors.

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Conflict of Interest

There are no conflicts of interest

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